RAM Acquisition

Introduction

In this practical you will use several tools to perform data acquisition on live computer systems (**Windows based**).

By the end of this seminar you will be able to

- Perform data acquisition by using Sysinternal tools
- Develop and improve your understanding and knowledge about RAM acquisition

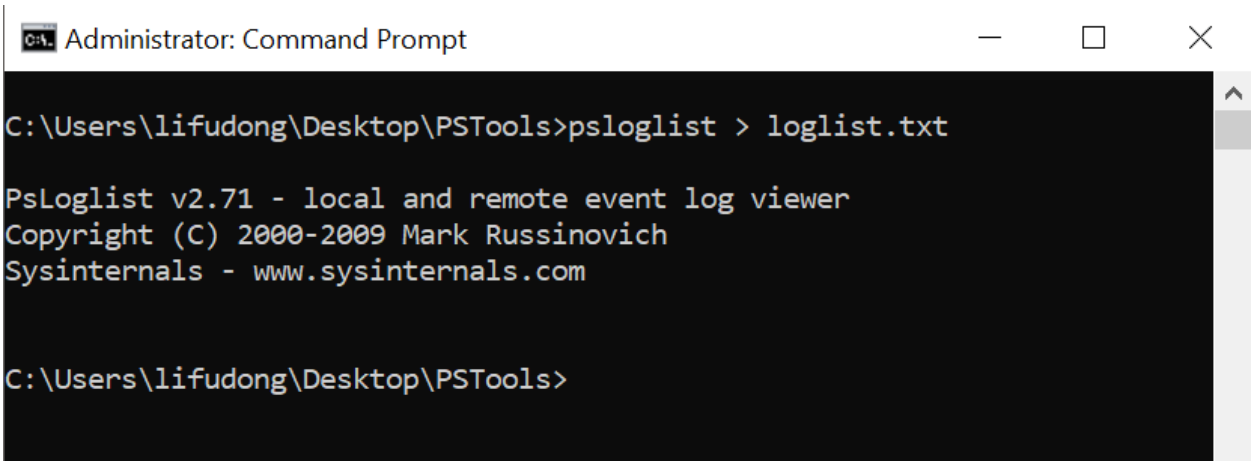
Task 1: Information gathering on live systems

For this task, you need to download the PsTools from the following link https://docs.microsoft.com/en - us/sysinternals/downloads/pstools ; ***alternatively***, you can visit the following links for Sysinternals Suite which contains more tools: https://docs.microsoft.com/en - us/sysinternals/downloads/sysinternals - suite or https://live.sysinternals.com/

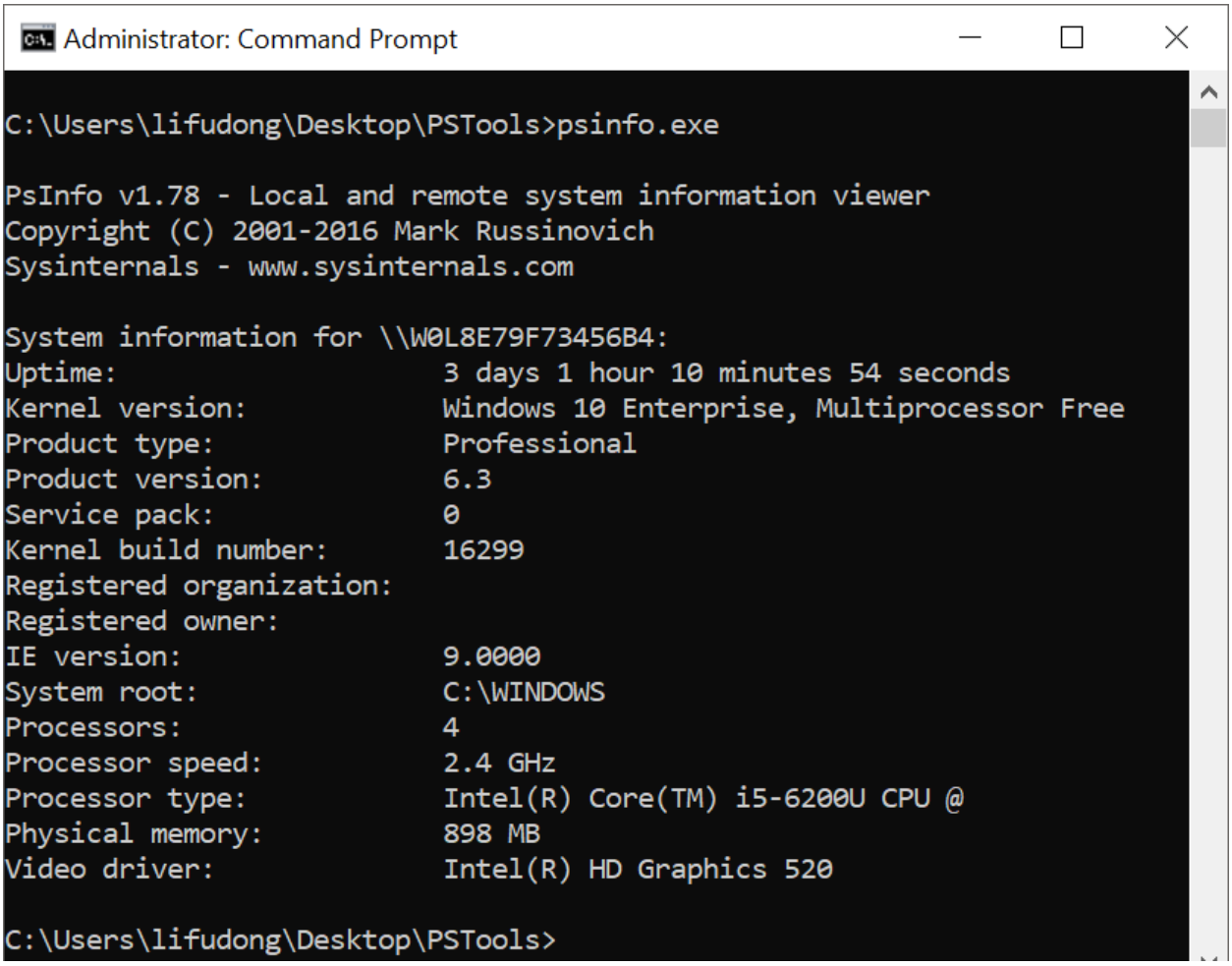
Use the Windows command prompt (probably run as administrator) and find out the information returned by using the following tools (hint: you need to navigate to the folder where the PsTools are saved/located first; then you will be able to use them.)

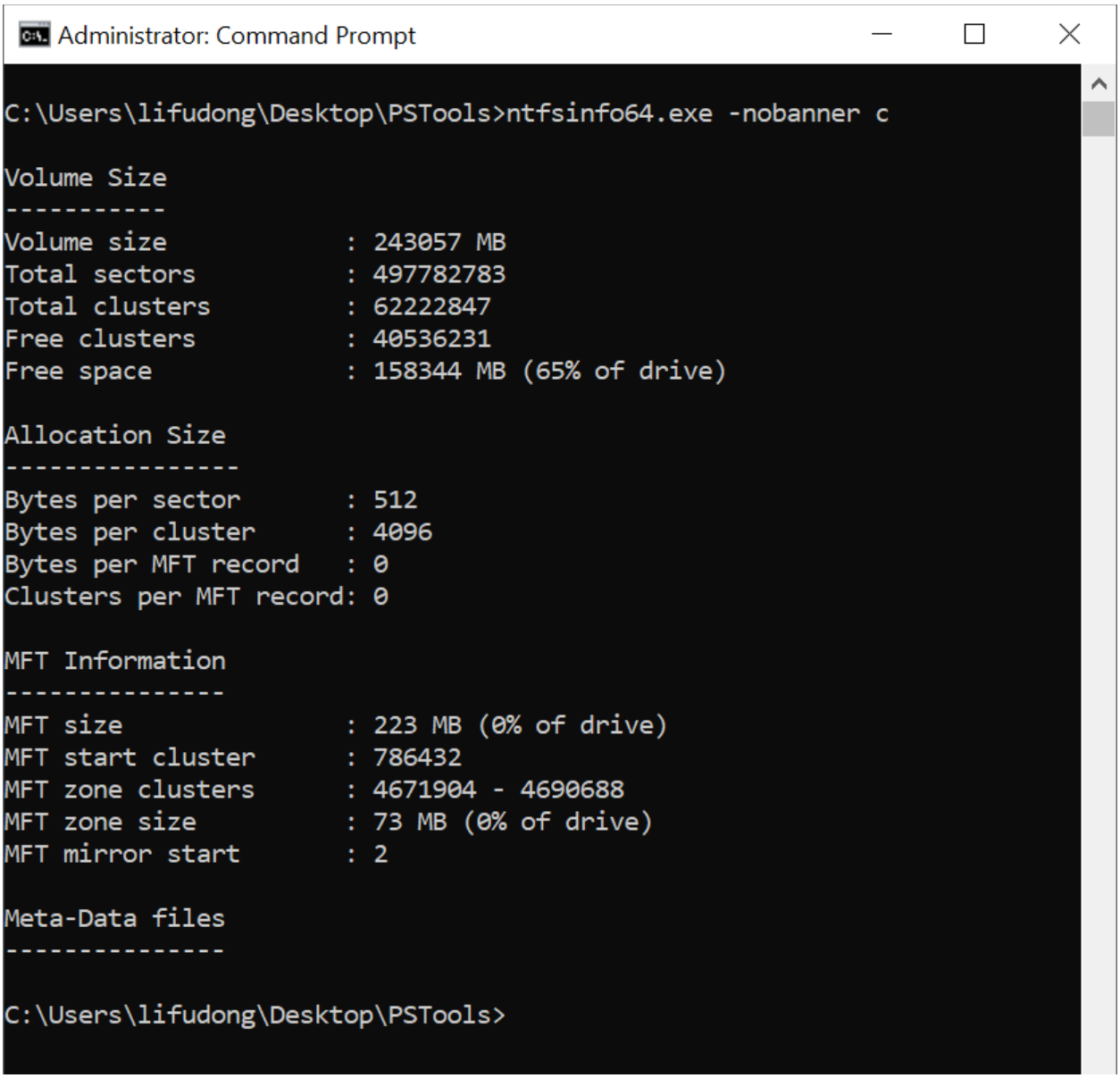
- psinfo.exe
- pslist.exe
- psfile.exe
- psservice.exe
- psloggedon.exe
- psloglist.exe
- hostname.exe
- logonsessions.exe
- ntfsinfo.exe
- procmon.exe

A few examples:



The output of the psloglist can be found in the loglist.txt file





You can try other tools that are available within the Sysinternals Suite.

Restart your computer and carry out Task 1 again and compare the difference between the two observations.

Task 2: Capture user password from the live memory

For this task, you need to download the User Mode Process Dumper Tool (userdump) (Version 8.1) from the following link

https://drive.google.com/file/d/1Ily9o9S_j0gfeUmpjXVJrPmFAuCyffrM/view?usp=drive_link

Use one of popular web browsers (e.g. Firefox) to login to one of your Internet services (e.g. University email). Then use the ***pslist*** (from Task 1) to identify the process id (PID) of the running browser. Use the userdump tool to capture the process related data from the memory and save it to a dmp file by using the following command [userdump 1780(pid)]. Check the following screenshot s (i.e. Figures 1 and 2) for more help if needed.

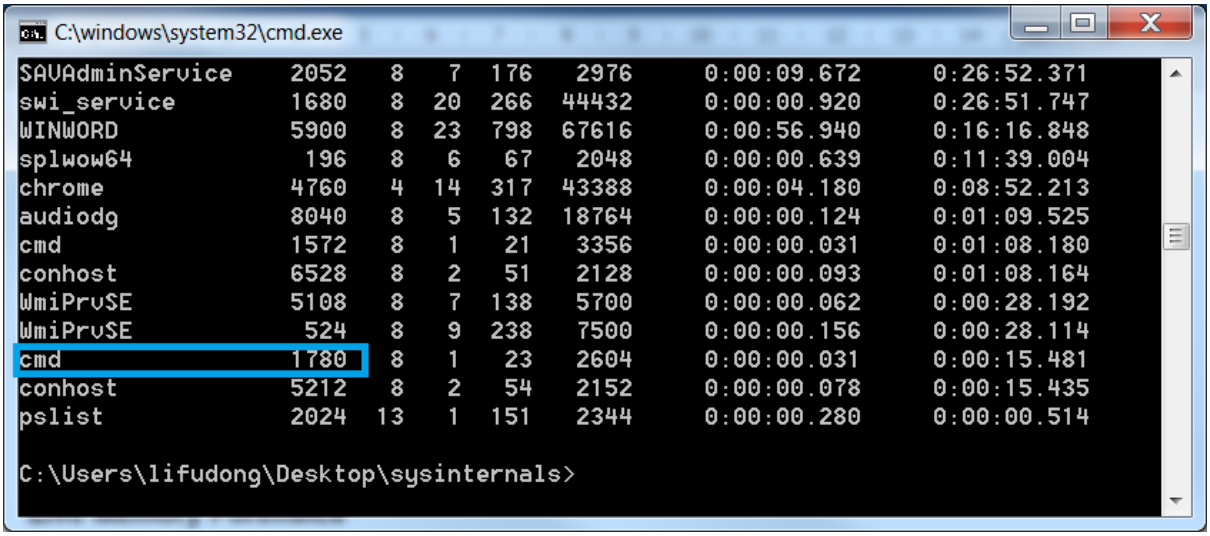


Figure 1: Using Pslist to capture process IDs **[There could be multiple running processes that are associated with a particular application; all of the running processes for an application need to be captured; rename the file dump before capturing the second process of the same application; otherwise t he file dump of the first running process of the application will be overwritten]**

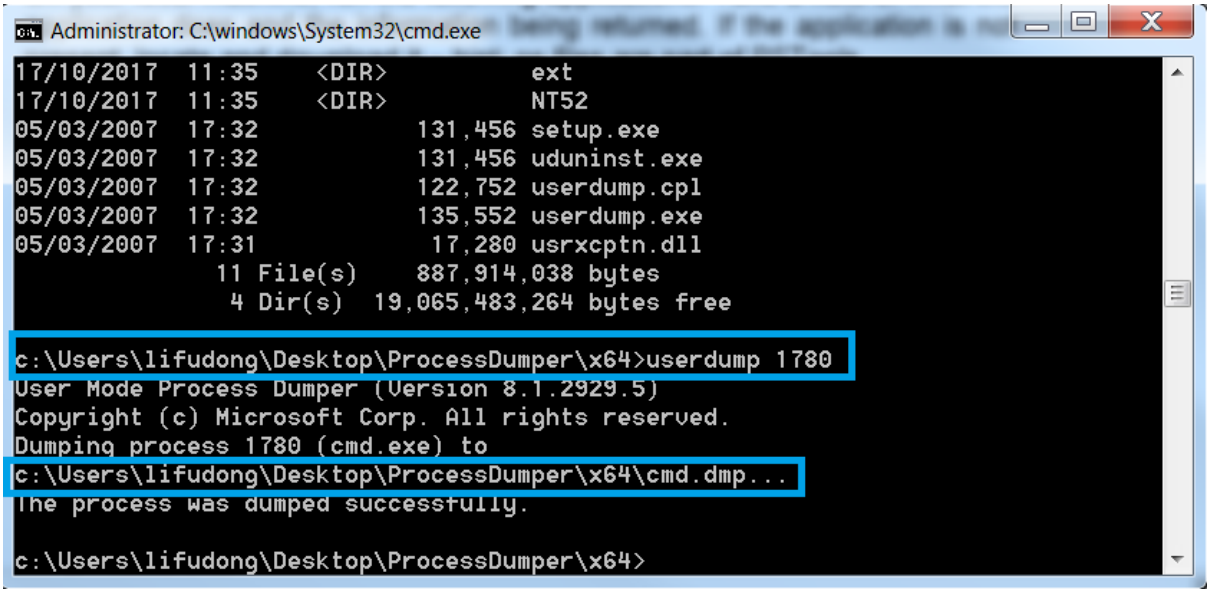


Figure 2: Using the userdump tool to capture the memory of process 1780

Use Autopsy to open the memory dump file and check whether you can find your online account password in plain text. (hint 1: choose the “Contents of A Folder” option when selecting the source evidence type; hint 2: make sure that no one is looking at your monitor when you are doing this task)

Task 3: Capture encryption password from the live memory

For this task, you need to follow the steps below:

1. Use 7-zip to encrypt a file of your choice by using a ***strong*** password; so you will have two files: one in plain text and one is encrypted

2. Delete the original file from step 1 (i.e. the plain text file), and ***ONLY*** keep the newly created .7z file

3. Open the .7z file via 7-zip and enter the password to view the file (***keep the file open until step 6***)

4. Use the pslist tool to identify the running process ID (PID) of the 7-zip program

5. Use the userdump tool to capture the 7-zip program data from the live memory and save it to a dmp file (***hint check Task 2 if needed***)

6. Use AccessData Imager to open the dump file and check whether you can find the password for your 7-zip encrypted file?